CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 12 : Python Program for Tic Tac Toe Game

Aim

To develop a Python program to play the Tic Tac Toe game between two players, X and O, and to find the winner or a draw.

Algorithm

- Create a board as a list of 9 empty cells and define a function to display it.
- Players X and O take turns, starting with X, and enter a position from 1 to 9.
- If the chosen cell is empty, place the player's mark; otherwise display "Position already taken".
- After each move, check all 8 winning combinations (3 rows, 3 columns, 2 diagonals).
- If a player has a winning combination, display the board, print the winner and stop.
- If all 9 moves are played with no winner, display the board and print "Draw!".

Program

```
board = [" "] * 9

def display():
    print(board[0], "|", board[1], "|", board[2])
    print("--+---+--")
    print(board[3], "|", board[4], "|", board[5])
    print("--+---+--")
    print(board[6], "|", board[7], "|", board[8])

for i in range(9):
    display()
    player = "X" if i % 2 == 0 else "O"
    pos = int(input("Player " + player + ", enter position (1-9): ")) - 1

    if board[pos] == " ":
        board[pos] = player
    else:
        print("Position already taken")
        continue

    if any(
        board[a] == board[b] == board[c] == player
        for a, b, c in [
            (0, 1, 2), (3, 4, 5), (6, 7, 8),
            (0, 3, 6), (1, 4, 7), (2, 5, 8),
            (0, 4, 8), (2, 4, 6)
        ]
    ):
        display()
        print("Player", player, "wins!")
        break
else:
    display()
    print("Draw!")
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=============== RESTART: C:/Users/shaik/OneDrive/Documents/12.py ===============
  |   |
--+---+--
  |   |
--+---+--
  |   |
Player X, enter position (1-9): 1
X |   |
--+---+--
  |   |
--+---+--
  |   |
Player O, enter position (1-9): 4
X |   |
--+---+--
O |   |
--+---+--
  |   |
Player X, enter position (1-9): 2
X | X |
--+---+--
O |   |
--+---+--
  |   |
Player O, enter position (1-9): 5
X | X |
--+---+--
O | O |
--+---+--
  |   |
Player X, enter position (1-9): 3
X | X | X
--+---+--
O | O |
--+---+--
  |   |
Player X wins!
>>>
```

RESULT

Thus, the Tic Tac Toe game was implemented successfully using Python.